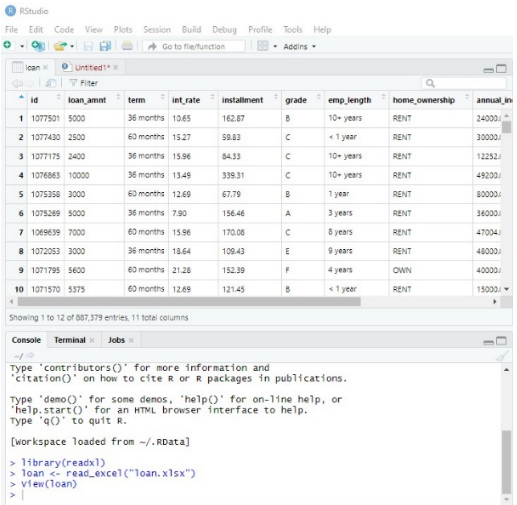
Week 4 Sean Graham

1.

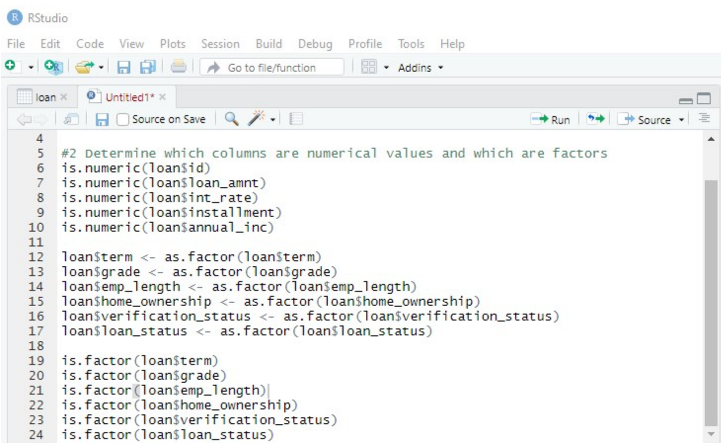


	id	loan_amnt	term	int_rate	installment	grade	emp_length	home_ownership	annual_in
1	1077501	5000	36 months	10.65	162.87	B	10+ years	RENT	24000.
2	1077430	2500	60 months	15.27	59.83	C	< 1 year	RENT	30000.
3	1077175	2400	36 months	15.96	84.33	C	10+ years	RENT	12252.
4	1076863	10000	36 months	13.49	339.31	C	10+ years	RENT	49200.
5	1075358	3000	60 months	12.69	67.79	B	1 year	RENT	80000.
6	1075269	5000	36 months	7.90	156.46	A	3 years	RENT	36000.
7	1069639	7000	60 months	15.96	170.08	C	8 years	RENT	47004.
8	1072053	3000	36 months	18.64	109.43	E	9 years	RENT	48000.
9	1071795	5600	60 months	21.28	152.39	F	4 years	OWN	40000.
10	1071570	5375	60 months	12.69	121.45	B	< 1 year	RENT	15000.

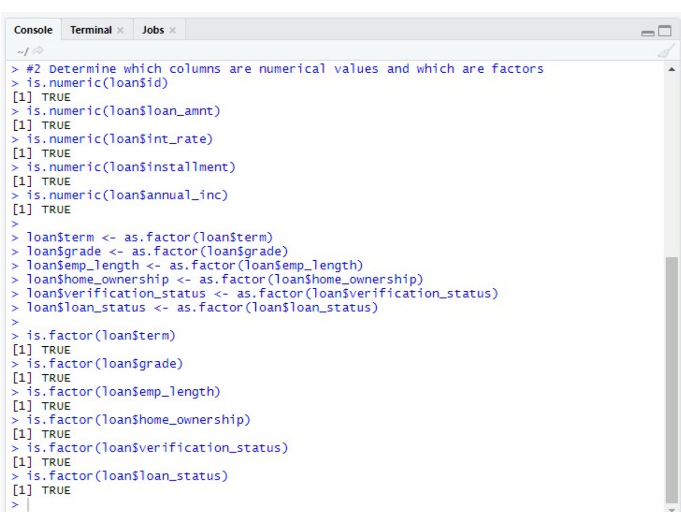
Showing 1 to 12 of 887,379 entries, 11 total columns

```
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

> library(readxl)
> loan <- read_excel("loan.xlsx")
> View(loan)
>
```

2. Id, loan_amnt, int_rate, installment, and annual_inc are numerical values. All the others are factors (once they are manually converted).

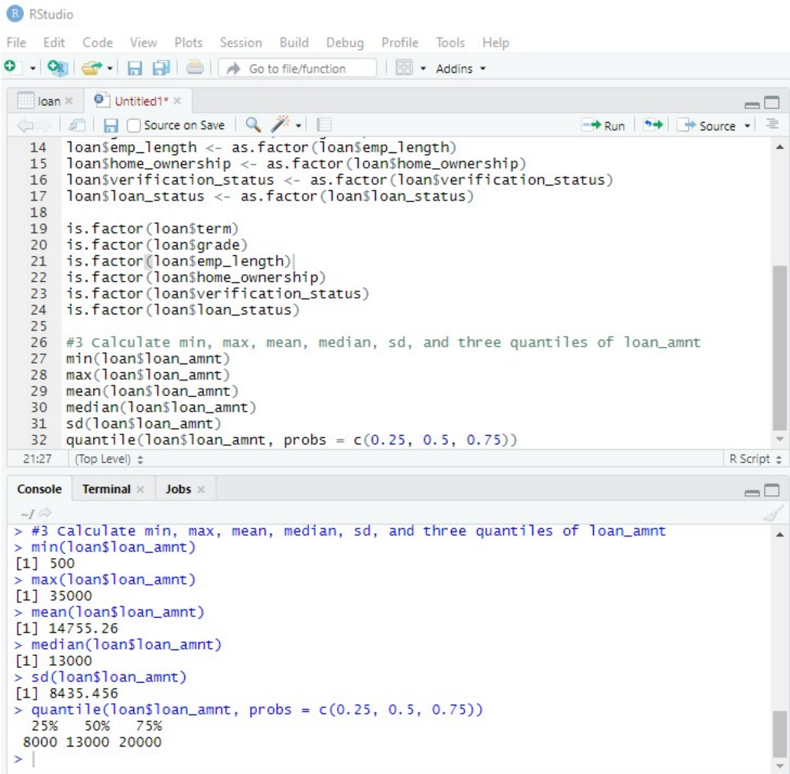


```
#2 Determine which columns are numerical values and which are factors
is.numeric(loan$id)
is.numeric(loan$loan_amnt)
is.numeric(loan$int_rate)
is.numeric(loan$installment)
is.numeric(loan$annual_inc)

loan$term <- as.factor(loan$term)
loan$grade <- as.factor(loan$grade)
loan$emp_length <- as.factor(loan$emp_length)
loan$home_ownership <- as.factor(loan$home_ownership)
loan$verification_status <- as.factor(loan$verification_status)
loan$loan_status <- as.factor(loan$loan_status)

is.factor(loan$term)
is.factor(loan$grade)
is.factor(loan$emp_length)
is.factor(loan$home_ownership)
is.factor(loan$verification_status)
is.factor(loan$loan_status)
```

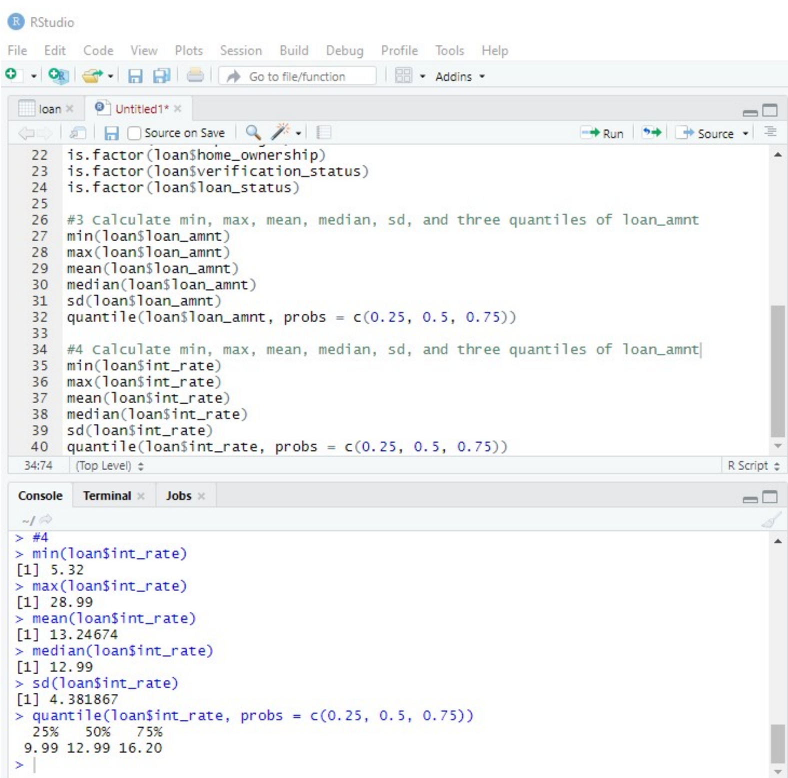


```
> #2 Determine which columns are numerical values and which are factors
> is.numeric(loan$id)
[1] TRUE
> is.numeric(loan$loan_amnt)
[1] TRUE
> is.numeric(loan$int_rate)
[1] TRUE
> is.numeric(loan$installment)
[1] TRUE
> is.numeric(loan$annual_inc)
[1] TRUE
>
> loan$term <- as.factor(loan$term)
> loan$grade <- as.factor(loan$grade)
> loan$emp_length <- as.factor(loan$emp_length)
> loan$home_ownership <- as.factor(loan$home_ownership)
> loan$verification_status <- as.factor(loan$verification_status)
> loan$loan_status <- as.factor(loan$loan_status)
>
> is.factor(loan$term)
[1] TRUE
> is.factor(loan$grade)
[1] TRUE
> is.factor(loan$emp_length)
[1] TRUE
> is.factor(loan$home_ownership)
[1] TRUE
> is.factor(loan$verification_status)
[1] TRUE
> is.factor(loan$loan_status)
[1] TRUE
>
```

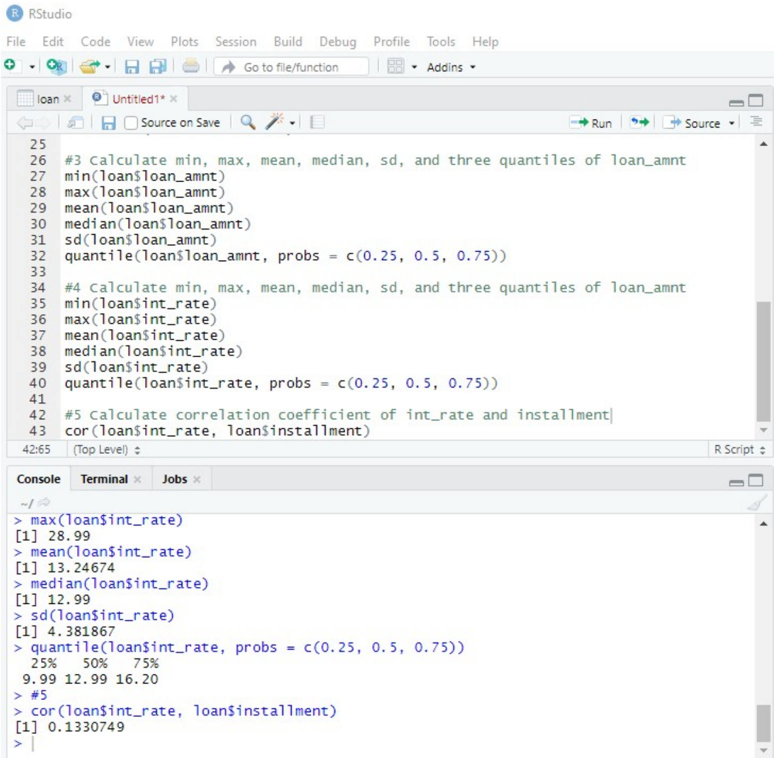
3.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
loan
Untitled1*
Source on Save
Run
Source
loan$emp_length <- as.factor(loan$emp_length)
loan$home_ownership <- as.factor(loan$home_ownership)
loan$verification_status <- as.factor(loan$verification_status)
loan$loan_status <- as.factor(loan$loan_status)
is.factor(loan$term)
is.factor(loan$grade)
is.factor(loan$emp_length)
is.factor(loan$home_ownership)
is.factor(loan$verification_status)
is.factor(loan$loan_status)
#3 Calculate min, max, mean, median, sd, and three quantiles of loan_amnt
min(loan$loan_amnt)
max(loan$loan_amnt)
mean(loan$loan_amnt)
median(loan$loan_amnt)
sd(loan$loan_amnt)
quantile(loan$loan_amnt, probs = c(0.25, 0.5, 0.75))
21:27
(Top Level)
R Script
Console
Terminal
Jobs
~/
> #3 Calculate min, max, mean, median, sd, and three quantiles of loan_amnt
> min(loan$loan_amnt)
[1] 500
> max(loan$loan_amnt)
[1] 35000
> mean(loan$loan_amnt)
[1] 14755.26
> median(loan$loan_amnt)
[1] 13000
> sd(loan$loan_amnt)
[1] 8435.456
> quantile(loan$loan_amnt, probs = c(0.25, 0.5, 0.75))
25% 50% 75%
8000 13000 20000
>

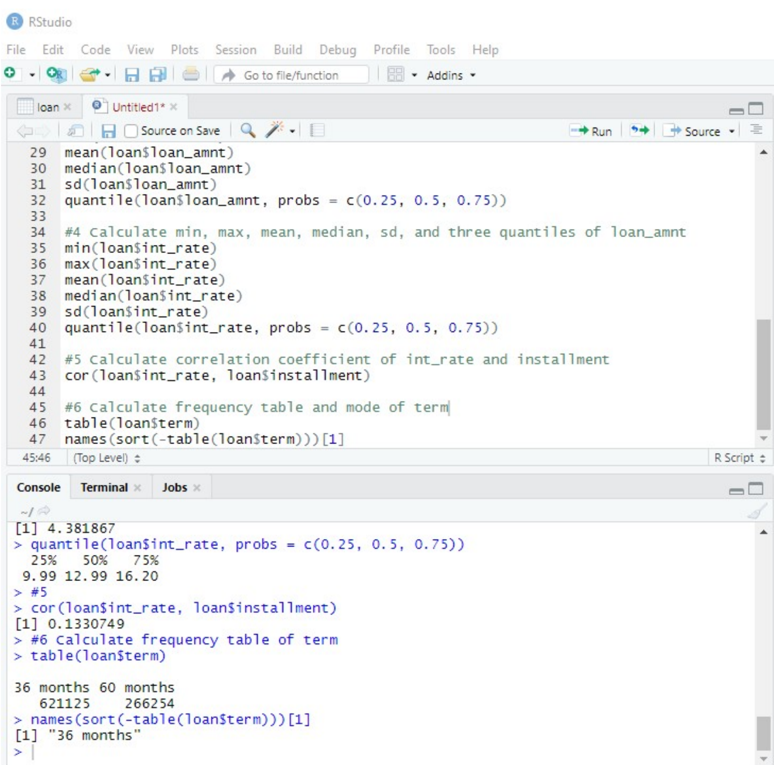
4.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
loan
Untitled1*
Source on Save
Run
Source
is.factor(loan$home_ownership)
is.factor(loan$verification_status)
is.factor(loan$loan_status)
#3 Calculate min, max, mean, median, sd, and three quantiles of loan_amnt
min(loan$loan_amnt)
max(loan$loan_amnt)
mean(loan$loan_amnt)
median(loan$loan_amnt)
sd(loan$loan_amnt)
quantile(loan$loan_amnt, probs = c(0.25, 0.5, 0.75))
#4 Calculate min, max, mean, median, sd, and three quantiles of loan_amnt
min(loan$int_rate)
max(loan$int_rate)
mean(loan$int_rate)
median(loan$int_rate)
sd(loan$int_rate)
quantile(loan$int_rate, probs = c(0.25, 0.5, 0.75))
34:74
(Top Level)
R Script
Console
Terminal
Jobs
~/
> #4
> min(loan$int_rate)
[1] 5.32
> max(loan$int_rate)
[1] 28.99
> mean(loan$int_rate)
[1] 13.24674
> median(loan$int_rate)
[1] 12.99
> sd(loan$int_rate)
[1] 4.381867
> quantile(loan$int_rate, probs = c(0.25, 0.5, 0.75))
25% 50% 75%
9.99 12.99 16.20
>

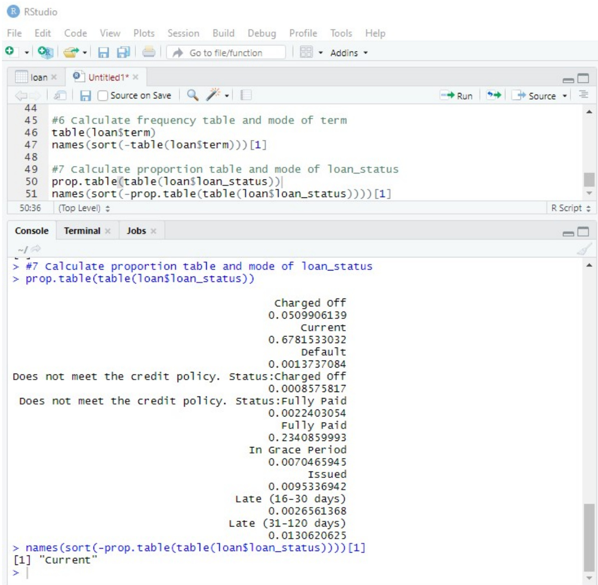
5. The correlation coefficient is close to 0, so the two variables do not have a string relationship.



6. The mode is “36 months.”



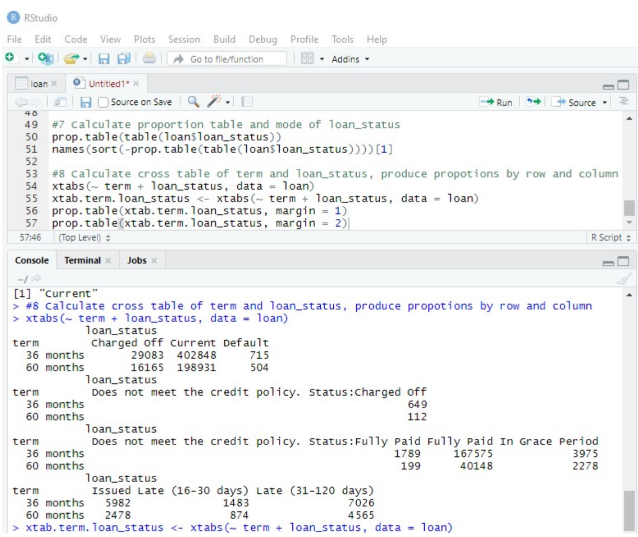
7. The mode is “Current”



```
#6 Calculate frequency table and mode of term
table(loan$term)
names(sort(-table(loan$term)))[1]

#7 Calculate proportion table and mode of loan_status
prop.table(table(loan$loan_status))
names(sort(-prop.table(table(loan$loan_status))))[1]
```

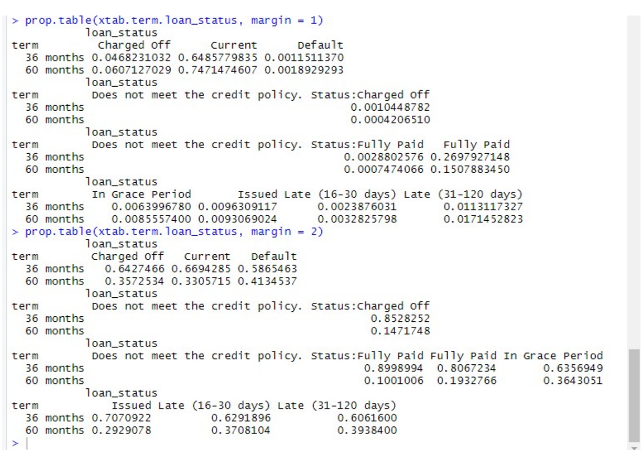
```
> #7 Calculate proportion table and mode of loan_status
> prop.table(table(loan$loan_status))

                                           Charged Off
                                          0.0509906139
                                               Current
                                          0.6781533032
                                               Default
                                          0.0013737084
Does not meet the credit policy. Status:Charged Off
                                          0.0008575817
 Does not meet the credit policy. Status:Fully Paid
                                          0.0022403054
                                            Fully Paid
                                          0.2340859993
                                       In Grace Period
                                          0.0070465945
                                                Issued
                                          0.0095336942
                                     Late (16-30 days)
                                          0.0026561368
                                    Late (31-120 days)
                                          0.0130620625
> names(sort(-prop.table(table(loan$loan_status))))[1]
[1] "Current"
>
```

8.



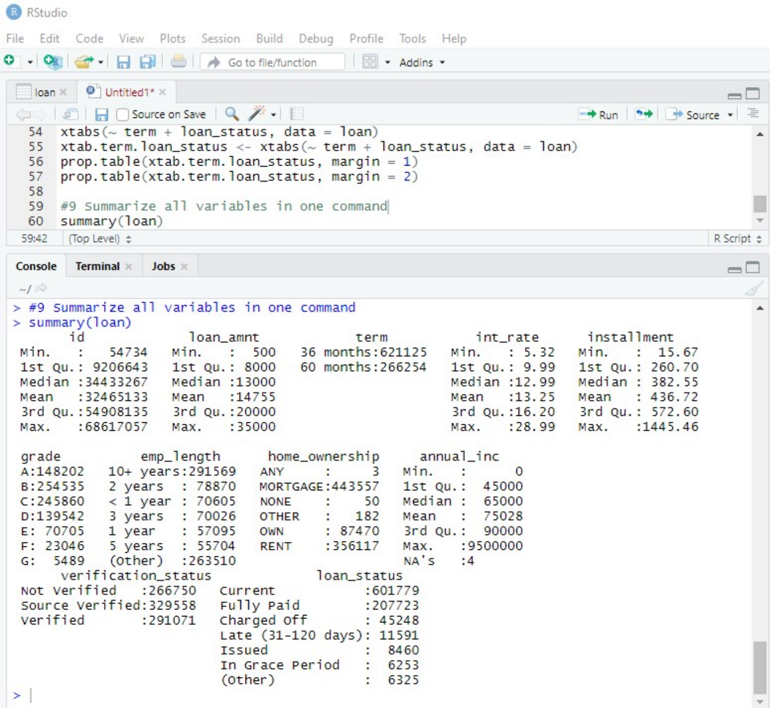
```
#7 Calculate proportion table and mode of loan_status
prop.table(table(loan$loan_status))
names(sort(-prop.table(table(loan$loan_status))))[1]

#8 Calculate cross table of term and loan_status, produce propotions by row and column
xtabs(~ term + loan_status, data = loan)
xtab.term.loan_status <- xtabs(~ term + loan_status, data = loan)
prop.table(xtab.term.loan_status, margin = 1)
prop.table(xtab.term.loan_status, margin = 2)
```

```
[1] "Current"
> #8 Calculate cross table of term and loan_status, produce propotions by row and column
> xtabs(~ term + loan_status, data = loan)
           loan_status
term         Charged Off Current Default
  36 months        29083  402848     715
  60 months        16165  198931     504
           loan_status
term         Does not meet the credit policy. Status:Charged Off
  36 months                                                  649
  60 months                                                  112
           loan_status
term         Does not meet the credit policy. Status:Fully Paid Fully Paid In Grace Period
  36 months                                                1789     167575            3975
  60 months                                                 199      40148            2278
           loan_status
term         Issued Late (16-30 days) Late (31-120 days)
  36 months    5982              1483               7026
  60 months    2478               874               4565
> xtab.term.loan_status <- xtabs(~ term + loan_status, data = loan)
```



```
> prop.table(xtab.term.loan_status, margin = 1)
           loan_status
term          Charged Off      Current      Default
  36 months 0.0468231032 0.6485779835 0.0011511370
  60 months 0.0607127029 0.7471474607 0.0018929293
           loan_status
term         Does not meet the credit policy. Status:Charged Off
  36 months                                         0.0010448782
  60 months                                         0.0004206510
           loan_status
term         Does not meet the credit policy. Status:Fully Paid   Fully Paid
  36 months                                        0.0028802576 0.2697927148
  60 months                                        0.0007474066 0.1507883450
           loan_status
term         In Grace Period       Issued Late (16-30 days) Late (31-120 days)
  36 months     0.0063996780 0.0096309117      0.0023876031       0.0113117327
  60 months     0.0085557400 0.0093069024      0.0032825798       0.0171452823
> prop.table(xtab.term.loan_status, margin = 2)
           loan_status
term         Charged Off   Current   Default
  36 months    0.6427466 0.6694285 0.5865463
  60 months    0.3572534 0.3305715 0.4134537
           loan_status
term         Does not meet the credit policy. Status:Charged Off
  36 months                                            0.8528252
  60 months                                            0.1471748
           loan_status
term         Does not meet the credit policy. Status:Fully Paid Fully Paid In Grace Period
  36 months                                           0.8998994  0.8067234       0.6356949
  60 months                                           0.1001006  0.1932766       0.3643051
           loan_status
term            Issued Late (16-30 days) Late (31-120 days)
  36 months 0.7070922         0.6291896          0.6061600
  60 months 0.2929078         0.3708104          0.3938400
>
```

9.



```
54  xtabs(~ term + loan_status, data = loan)
55  xtab.term.loan_status <- xtabs(~ term + loan_status, data = loan)
56  prop.table(xtab.term.loan_status, margin = 1)
57  prop.table(xtab.term.loan_status, margin = 2)
58
59  #9 Summarize all variables in one command
60  summary(loan)
```

```
> #9 Summarize all variables in one command
> summary(loan)
       id             loan_amnt            term           int_rate       installment
 Min.   :   54734   Min.   :  500   36 months:621125   Min.   : 5.32   Min.   :  15.67
 1st Qu.: 9206643   1st Qu.: 8000   60 months:266254   1st Qu.: 9.99   1st Qu.: 260.70
 Median :34433267   Median :13000                      Median :12.99   Median : 382.55
 Mean   :32465133   Mean   :14755                      Mean   :13.25   Mean   : 436.72
 3rd Qu.:54908135   3rd Qu.:20000                      3rd Qu.:16.20   3rd Qu.: 572.60
 Max.   :68617057   Max.   :35000                      Max.   :28.99   Max.   :1445.46

 grade          emp_length      home_ownership     annual_inc
 A:148202   10+ years:291569   ANY     :     3   Min.   :      0
 B:254535   2 years  : 78870   MORTGAGE:443557   1st Qu.:  45000
 C:245860   < 1 year : 70605   NONE    :    50   Median :  65000
 D:139542   3 years  : 70026   OTHER   :   182   Mean   :  75028
 E: 70705   1 year   : 57095   OWN     : 87470   3rd Qu.:  90000
 F: 23046   5 years  : 55704   RENT    :356117   Max.   :9500000
 G:  5489   (Other)  :263510                     NA's   :4
      verification_status               loan_status
 Not Verified   :266750   Current              :601779
 Source Verified:329558   Fully Paid           :207723
 Verified       :291071   Charged Off          : 45248
                          Late (31-120 days): 11591
                          Issued               :  8460
                          In Grace Period      :  6253
                          (Other)              :  6325
> 
```